ADAMAS: an open, tested framework for diamond electronics and room-temperature nitrogen-vacancy quantum processors

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github.com/Normansrule/adamas-diamond-framework

Preprint, generated from the repository

Abstract

Diamond combines the largest breakdown field and thermal conductivity of any practical semiconductor with a defect, the nitrogen-vacancy (NV) center, whose electron spin is a qubit at room temperature. ADAMAS is an open framework that follows one path from wafer to quantum processor with executable, cited models: 583 references (500 verified against Crossref), 53 regenerable figures, a process design kit with a placed 4-bit processor, circuit-level quantum error correction, and hands-on experiments from US$100 upward. An expert audit (five reviewer roles, 80 findings) corrected the earlier version; four results survive it. (i) Only 0.50% of the boron in a bulk-doped 1.2 kV drift is ionized at 300 K, so in an 800 V traction inverter a bulk-doped diamond switch loses to silicon carbide (loss ratio 0.40) and wins only when hot (1.9× at a 523 K junction). (ii) For the published room-temperature NV pair, coherence alone allows a Bell fidelity of 0.78 to 0.998; the measured 0.67 lies below that range, a discrepancy the models do not explain. (iii) Circuit-level simulations with idle errors during millisecond readout show that surface-code scaling survives only while readout stays well below the nuclear-memory coherence time; a bias-tailored XZZX code extends this about tenfold, and with realistic two-level gates (bias on CZ only) it doubles the threshold to 1.81%. (iv) A factoring-scale machine calibrated to the published 8-hour superconducting estimate fits on a 5.5 mm die but runs 2.1 years at 1 ms readout, assuming an inter-cell link that has not been demonstrated. Every number in this paper is generated from the package.

1 Introduction

Silicon's dominance rests on process maturity, not material merit. For power conversion, high-temperature electronics, and room-temperature quantum technology, diamond's figures of merit are extreme: its Baliga figure of merit for power switches is 25,777× silicon's (holes at 2000 $\mathrm{cm}^2/\mathrm{V\,s}$ and 10 MV/cm; a smaller value at the 7.7 MV/cm measured on small samples) and its thermal conductivity 14.7× [1, 2, 3]. Its obstacles are equally concrete: boron acceptors sit 0.37 eV deep, so only 0.50% of the boron in a 1.2 kV drift ionizes at room temperature [4]; the largest peer-reviewed single-crystal wafers are 2-inch heteroepitaxial plates with about 2×10^7 dislocations per cm^2 [5, 6], on which the critical field falls to 1–2 MV/cm [7, 8] (a 3-inch process has been announced [9]); and logic was p-channel only until an n-channel MOSFET reported in 2024 [10].

ADAMAS treats these facts as inputs to models that can be run, tested, and falsified. The contribution is not a new device but a connected, reproducible chain: materials → process → circuits → qubits → error correction → machine size → experiments, with each link checked against published data or an independent tool (ngspice, Icarus Verilog, Yosys, KLayout, Stim, PyMatching).

2 Power devices and converters

The unipolar limit $R_{on,sp} = 4BV^2/(\varepsilon\mu E_c^3)$ [1] places ideal diamond far below every other material (Fig. 1). Measured lateral hole-gas transistors reach 3659 V [13] and, at 2568 V, 7.54 $\mathrm{m\Omega\,cm}^2$ [12], about 89× below silicon's vertical limit (indicative only, since the devices are lateral). For an area-optimized hard-switched leg, conduction and switching losses equalize and the loss is $P = 2IV\sqrt{fR_{on,sp}C_{oss,sp}/2}$ [14]. A bulk boron-doped drift conducts only through its ionized fraction, so $R_{on}(T) = R_{ideal}\,(T/300)^{\alpha}/f_{ion}(T)$, with $f_{ion} = 0.50\%$ at 300 K and 14.8% at 523 K for the 1.2 kV design (uncompensated). Table 1 shows the result for an 800 V, 300 A, 20 kHz traction inverter: diamond wins when hot

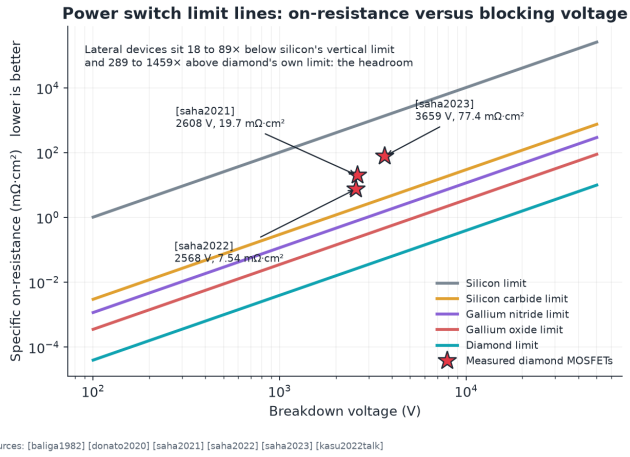


Figure 1: One-dimensional on-resistance limits and measured hydrogen-terminated diamond MOSFETs [11, 12, 13]. Real devices sit above diamond's own limit but below silicon's.

and loses at room temperature. Delta-doped and hole-gas channels avoid the ionization penalty (proposed here); the measured lateral device has none.

Switch, 1.2 kV (W)	300 K	523 K
Silicon, ideal	209.5	–
Silicon carbide, ideal	29.7	58.0
Gallium nitride, ideal	20.6	–
Diamond, bulk-doped drift	74.6	29.8
Diamond, 2022 lateral, scaled	89	–

Table 1: Area-optimized switch loss in an 800 V, 300 A, 20 kHz inverter leg (first order: no gate charge or reverse recovery). The bulk-doped diamond drift includes incomplete ionization of boron.

3 Process, design kit, and a placed processor

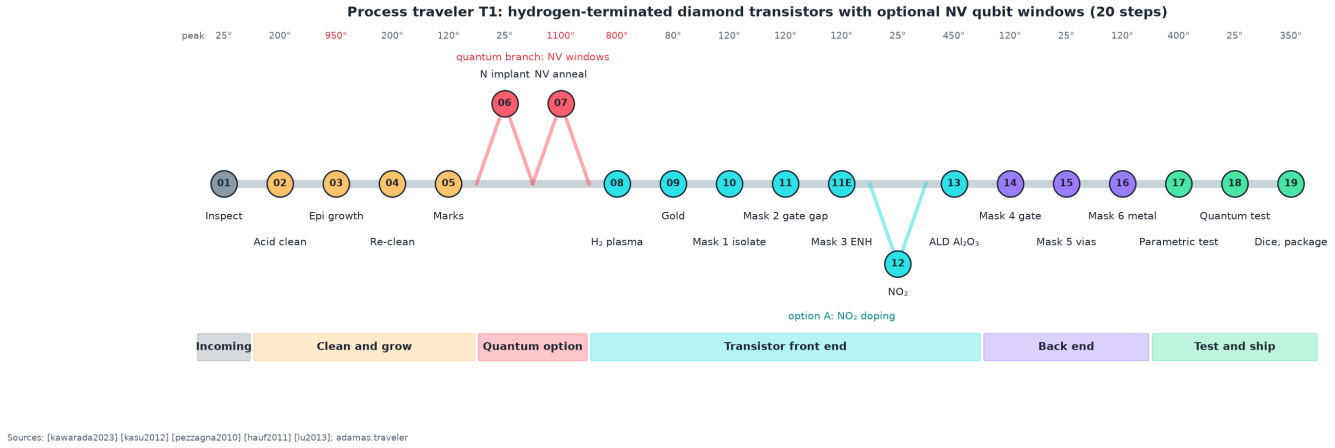
The hydrogen-terminated flow (Fig. 2) creates the channel by surface transfer doping [15], protects it with gold immediately, isolates devices with one oxygen-plasma step, and stabilizes the hole gas with atomic-layer-deposited Al_2O_3 at ≤150 °C (with NO_2) or ∼450 °C (for 400 °C operation) [16, 17, 18]. Hydrogen plasma at 700–1000 °C passivates NV centers [19], so wafers that already hold qubits take the low-temperature (≤500 °C) hydrogen option with the NV windows masked. Its thermal budget forces growth and the NV anneal (up to 1100 °C [20]) ahead of all metal. The traveler gives parameters with sources, pass criteria, safety, and failure modes for each step, and an interactive run log. On this process, the design kit PDK-0 supplies layers, rules, a GDS monitor die, standard cells, LEF abstracts, and a timed Liberty file; the DIA-4 processor synthesizes to 323 cells and places in about 1 mm^2 with pads. An emulator matches its Verilog cycle for cycle on random programs.

4 The two-qubit bottleneck

With the published pair's parameters, 25 nm spacing, 4.93 kHz single-quantum coupling (a 25 μs double-quantum gate), and echo times of 150 and 514 μs [21], the coherence-limited Bell fidelity is 0.78 to 0.998 over echo-decay exponents 1 to 3, one or two gate applications, and with or without T_1 (0.998 for the cubic exponent and one application). The measured 0.67, normalized to the charge-preselected scheme of [21], lies below this whole range. An earlier version of this framework read the gap as a charge-state preparation probability $q = 0.75$; because the measurement was already charge-preselected, that reading is withdrawn and the gap is recorded as a known discrepancy (Fig. 3). The two pairs were addressed spectrally (different crystal axes) and read out jointly; at 9.5 nm, two-qubit fidelity 96% has since been shown with 594 nm charge pre-selection [22]. A five-level rate model with measured rates [23] gives a readout contrast of 45% for a pure state in a 300 ns window and 82% spin polarization; starting from that imperfectly polarized state the measurable contrast is 31%, as observed.

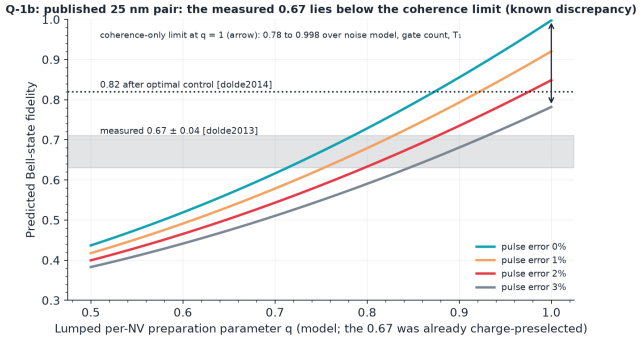
5 Error correction with slow readout

NV readout takes about a millisecond [24, 25], during which data qubits held in nuclear spins idle. Circuit-level simulations with Stim and PyMatching [26, 27] (0.3% link, 1% readout, 0.2% preparation errors) validate a threshold near 1.2% for uniform noise (the phenomenological value is 3.1%) and give an error-suppression factor $\Lambda = 4.03$ at 0.1 ms readout, 3.56 at 1 ms, and 2.64 at 10 ms for a 1 s memory (fitted on $d \leq 7$ and extrapolated). Distance 7 stops beating distance 3 near 100 ms; readout that at most doubles the distance-7 error must stay below about 1% of the memory time. Nuclear memories mostly dephase; the standard code wastes this bias because its X memory fails first, while an XZZX code [28, 29], emulated by Hadamard conjugation of a checkerboard sublattice, pushes break-even to about 1000 ms (Fig. 4). This idle-noise result needs no bias-preserving gate. Building the XZZX syndrome circuit natively on a CZ-native gate set includes hook errors; because a two-level qubit cannot have a bias-preserving CNOT [30, 31], CZ gates keep the bias η and CNOTs at most $\eta = 5$ [32]. At $\eta = 100$ the standard code's threshold stays near 0.86% (0.93% unbiased) while the XZZX threshold rises to 1.81% (3.25% would need bias-preserving CNOTs), and for an NV cell at 1 ms readout the distance-7 XZZX memory is 9× better than the standard one (Fig. 5).



Sources: [kawarada2023] [kasu2012] [pezzagna2010] [hauf2011] [lu2013]; adamas.traveler

Figure 2: Process traveler T1: 20 steps from a bare plate to a tested chip, with a quantum branch (NV implant and anneal) and an NO_2 doping option. Heat is spent before the first metal.

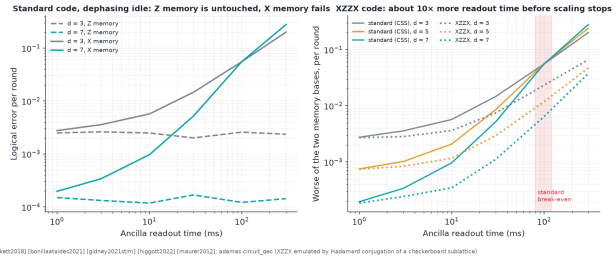


Sources: [dolde2013] [dolde2014]; adamas.gate_budget model (cubic echo exponent, one gate application for the curves)

Figure 3: Echoed NV–NV gate budget evaluated with the published pair [21]: fidelity versus preparation probability. Coherence alone allows 0.78 to 0.998 depending on the noise model; the measured 0.67 lies below.

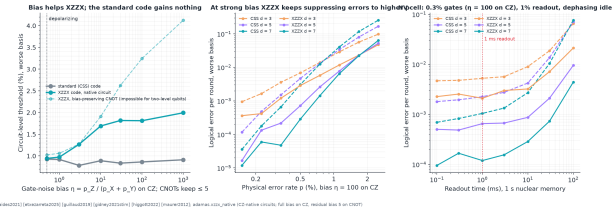
6 How big, how slow

Fitting $p_L(d) = A\Lambda^{-(d+1)/2}$ [33] and pricing each logical step at d rounds [34] gives, for RSA-2048 (6,000 logical qubits and 1.07×10^9 logical steps, the step count at which the published parameters of [35], distance 27 and a 1 μs cycle, give their published 8 hours), distance 55, 73 million physical qubits on a 5.5 mm die, and 2.1 years at 1 ms readout (0.3 years at 0.1 ms), against 30 hours for a superconducting machine at today's measured $\Lambda = 2.14$ [36] (Fig. 6). Counting each of the about 3×10^9 Toffoli gates (2.7×10^9, rounded up) as a serial step gives an upper bound of 6.0 years. Each round contains four inter-cell operations of 25 μs at 0.3% error: at the 1 μm cell pitch these need a mediated link (spin-chain bus or photonic), which has not been demonstrated. With biased gate noise (bias 100 on CZ, 5 on CNOT) the standard code needs distance 85 and 173 million qubits (3.2 years), while a native XZZX code needs distance 49, 58 million qubits, and 1.8 years. Once readout is fast, the link dominates each round.



Sources: [tuckett2018] [bonilla ataides2021] [gidney2021stim] [higgott2022] [maurer2012]; adamas.circuit_qec (XZZX emulated by Hadamard conjugation of a checkerboard sublattice)

Figure 4: Circuit-level surface code with dephasing idle errors during readout. Left: standard code, Z and X memories. Right: worse basis for standard and XZZX codes.



Sources: [bonilla ataides2021] [wieckowska2020] [gu-tbias2018] [gidney2021stim] [higgott2022] [maurer2012]; adamas.xzzx_native (CZ-native circuits; full bias on CZ, residual bias 5 on CNOT)

Figure 5: Native XZZX circuits with biased gate noise (full bias on CZ, at most 5 on CNOT): thresholds versus bias, logical error at $\eta = 100$, and the NV readout scenario.

7 Experiments and teaching

Ten experiments span US$100 to a pilot line (Fig. 7), following published teaching setups [37, 38, 39]. The cheapest ships with firmware (Raspberry Pi Pico driving an ADF4351 synthesizer) and a host script whose output feeds the same fitting code as a browser Data Lab. An eight-lesson course, 15 interactive labs, fourteen animated equations tied one-to-one to the experiments, and a process traveler turn the framework into teaching material, also packaged as an offline desktop app; all browser simulations are tested against the Python models.

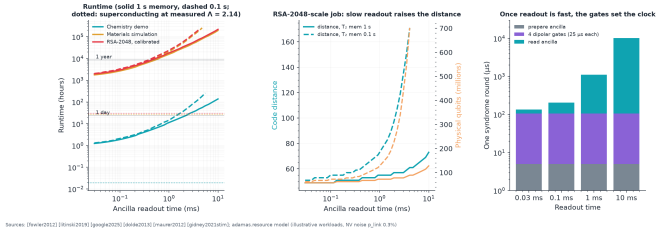


Figure 6: Resource estimate from the circuit-level fits: runtime, code distance, and the composition of one syndrome round.

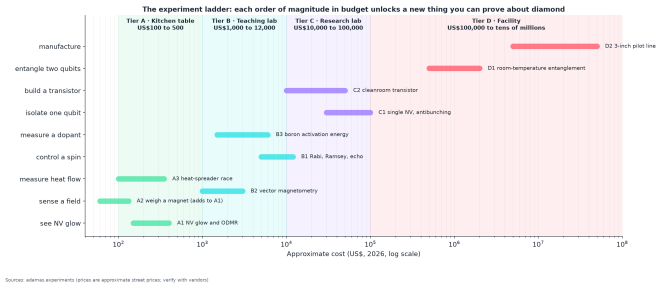


Figure 7: Ten experiments in four budgets, with parts, steps, safety, and simulated expected results; a firmware kit makes the US$150 to 400 ODMR experiment runnable.

8 Validation and uncertainty

A validation matrix of 18 checks (17 pass, the rest explained discrepancies) compares each model with a published result, an independent tool, or a known answer; the phenomenological surface-code threshold, for example, reproduces the published matching-decoder value of 2.93% to within 6% [40]. Propagating the published spread of diamond's critical field and hole mobility [41, 3, 42, 43] gives a silicon-carbide-to-diamond loss ratio with median 0.36 at 300 K (P10 to P90: 0.26 to 0.49) and 1.7 at 523 K. The runtime of the calibrated RSA-2048 machine spans 0.4 to 5.3 years (median 1.2), dominated by readout time (Fig. 8). Variance-based (Sobol) sensitivity indices [44, 45, 46] make the ranking quantitative: diamond's hole mobility and critical field explain about 0.64 and 0.17 of the power-ratio variance (a weakly ionized drift makes the ratio scale as $\sqrt{\mu E_c}$), and readout time 0.96 of the log runtime. Dividing by the cost of the experiment that pins each input down ranks a few-thousand-dollar Hall measurement of diamond hole mobility first; boron-nitride-gated hole channels near 1000 $\mathrm{cm}^2/\mathrm{V\,s}$ [47] show why the range is wide. Readout, the quantum side's dominant input, is improving through nuclear-assisted repetitive readout [48] and ionization-based initialization [49]. Finally, the XZZX code needs fewer qubits than the standard code whenever the gates are biased and nearly ties without bias (142 versus 135 million), so measuring the bias is worth only 2.4 million qubits for the code choice [50]: it sizes the machine more than it changes its code (Fig. 9).

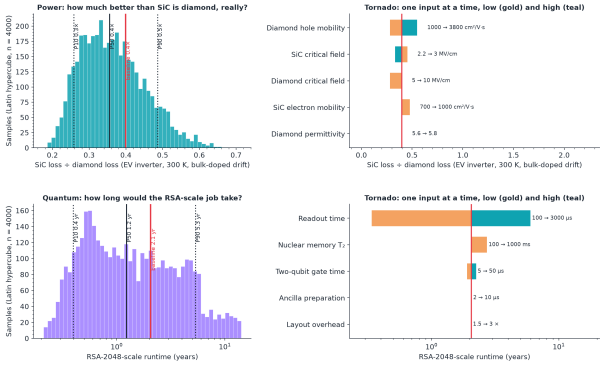


Figure 8: Latin-hypercube uncertainty propagation and tornado diagrams for the two headline results.

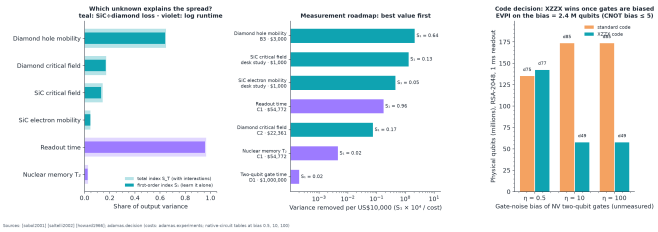


Figure 9: What to measure next: Sobol indices, variance removed per dollar, and the code decision under unknown gate bias.

9 Limitations

Transistor and switching parameters are literature-anchored placeholders until the PDK-0 monitor die is measured. The converter model omits gate charge and reverse recovery and assumes an uncompensated drift; the surface-code idle channel is phenomenological at the gate level (depolarizing or pure dephasing); the bias of NV two-qubit gates and the residual bias of a CNOT are assumptions; and the inter-cell link of the machine estimate is proposed, not demonstrated. The chemistry and materials workloads are illustrative scales; the RSA-2048 workload is calibrated to a published runtime.

Reproducibility

`pip install -e ".[dev]" && python -m pytest -q && make figures paper` regenerates every figure, the numbers in this text (`python -m adamas.paper`), and this PDF. Interactive version: normansrule.github.io/adamas-diamond-framework.

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